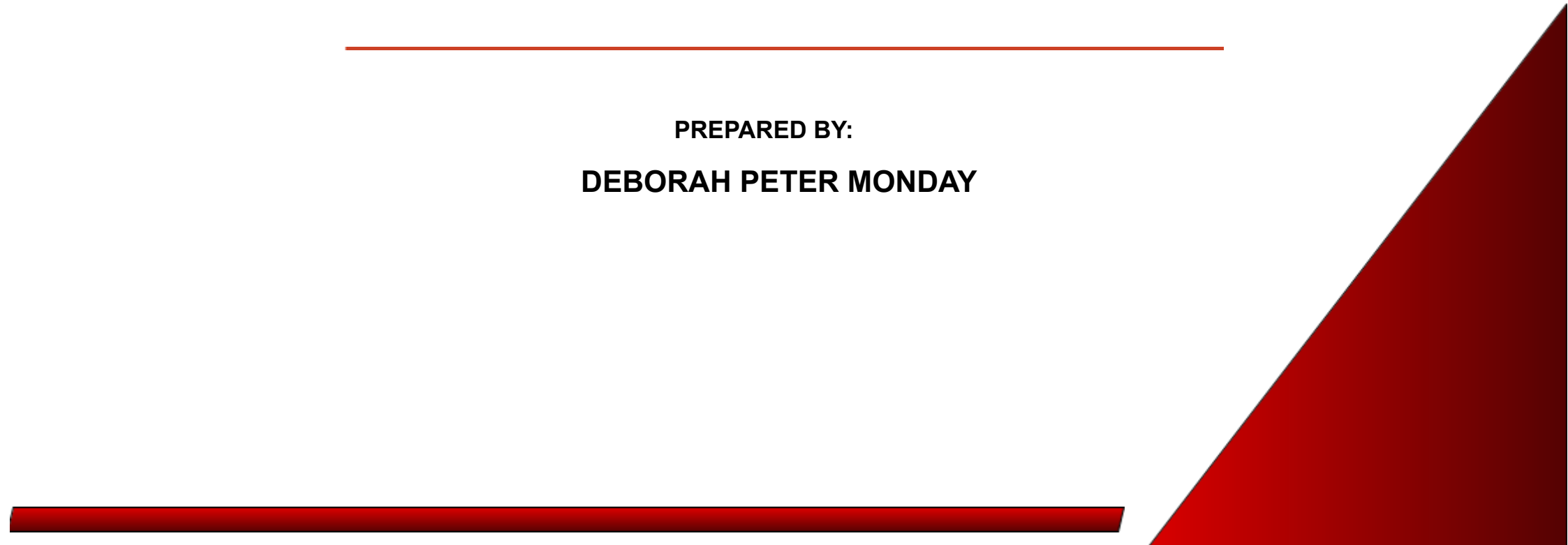


NETWORKING PROJECT

DESIGN AND IMPLEMENTATION OF A SINGLE LOCAL AREA NETWORK (LAN) USING DHCP

PREPARED BY:

DEBORAH PETER MONDAY

The slide features decorative red geometric elements: a large red triangle in the bottom right corner and a horizontal red bar at the bottom left.

Project outcome

Objectives:

The objective of this project is to design and implement a structured single Local Area Network that utilizes dynamic IP address allocation through DHCP, while validating seamless communication between connected devices.

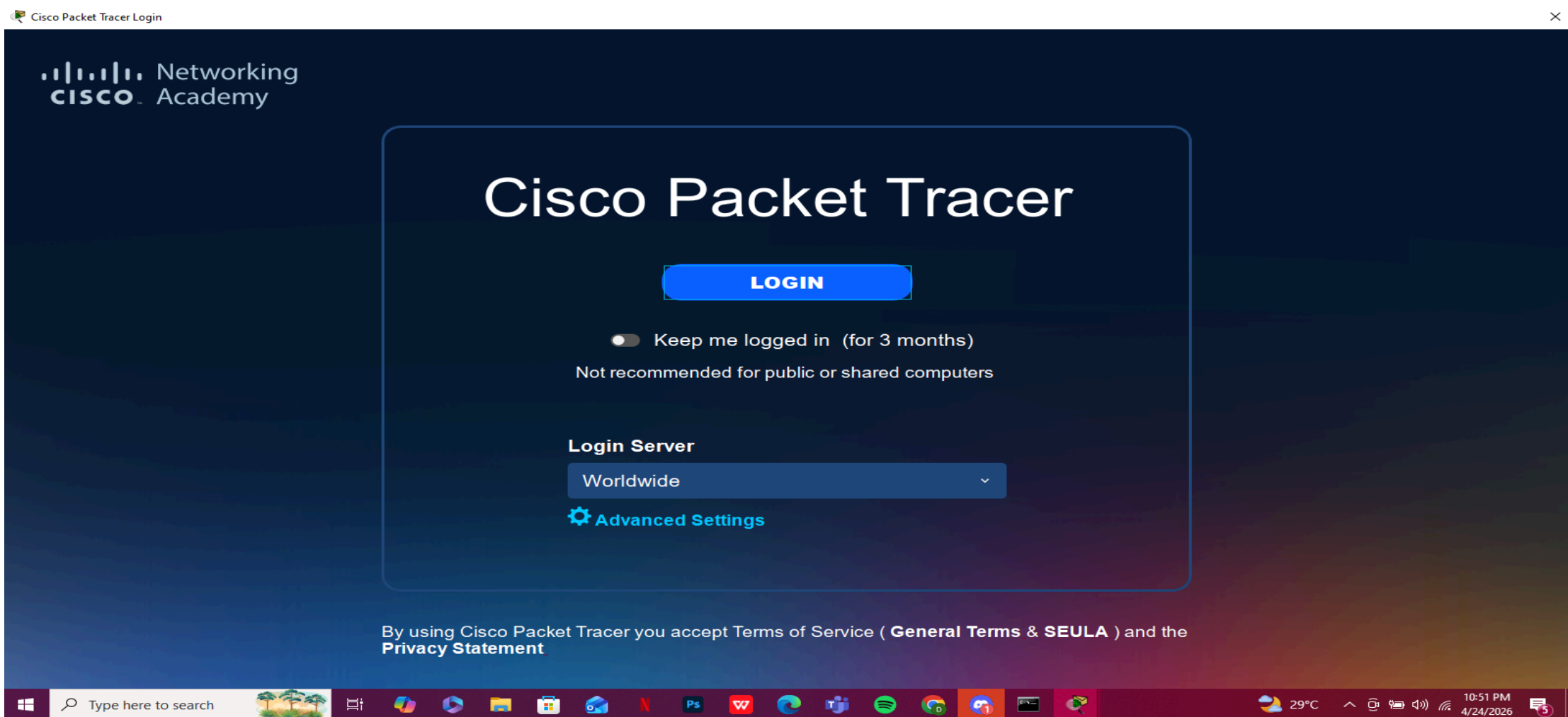
Key components used in this project:

Component	Description
• CISCO Packet Tracer	Used for designing and simulating the network
• DHCP server	provides automatic IP address allocation
• Switch	Connect all devices within the LAN
• PCs & Laptop	End devices that receive IP addresses and communicate.
• IP Addressing (172.50.0.0/16)	Defines the network and enables communication
• Cable	Used to physically connect devices

Step 1:

Launching and Accessing Cisco Packet Tracer

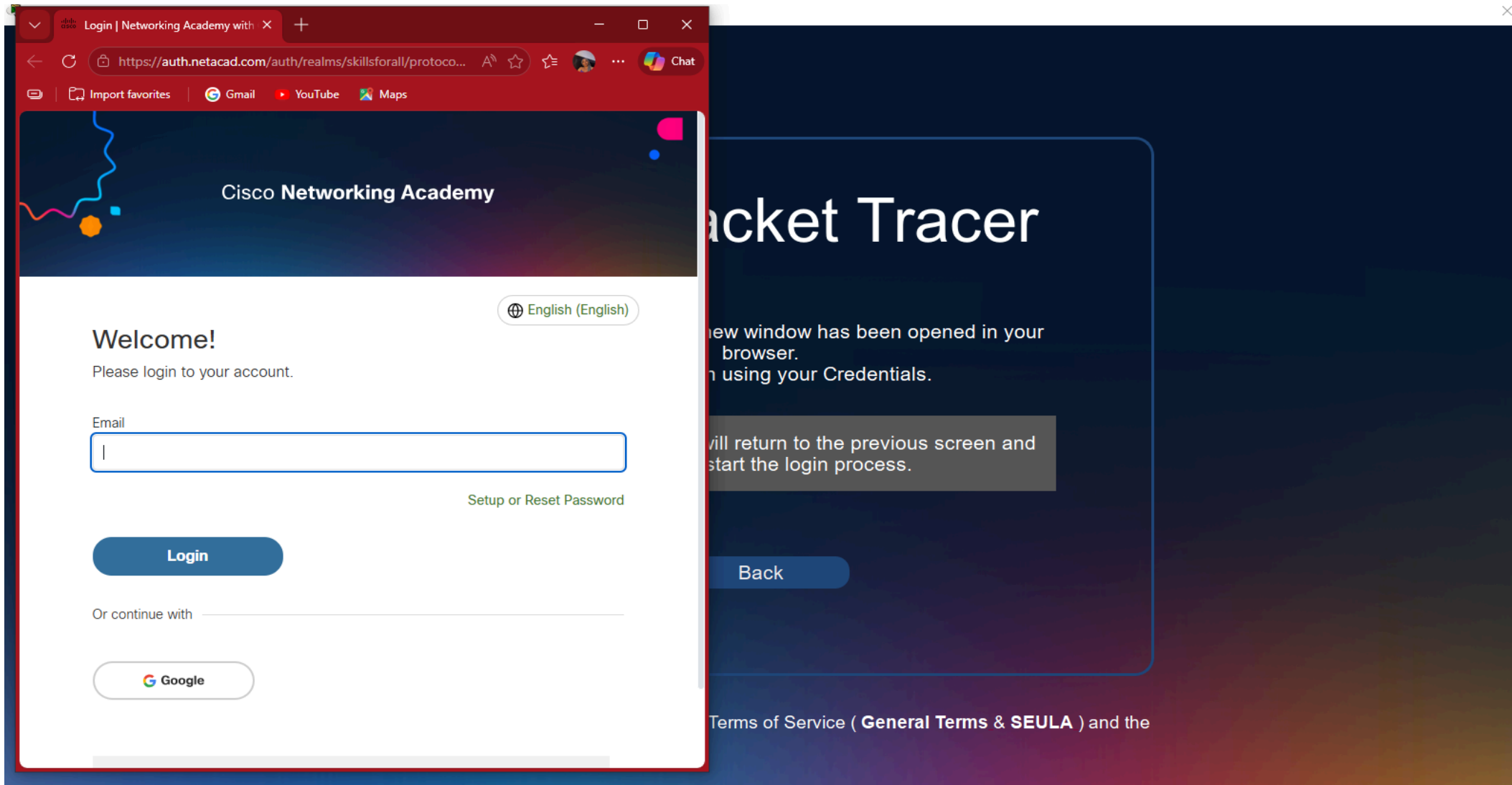
Click on the login button and wait for the authentication page to load before proceeding.



Step 2:

User Authentication in Cisco Packet Tracer

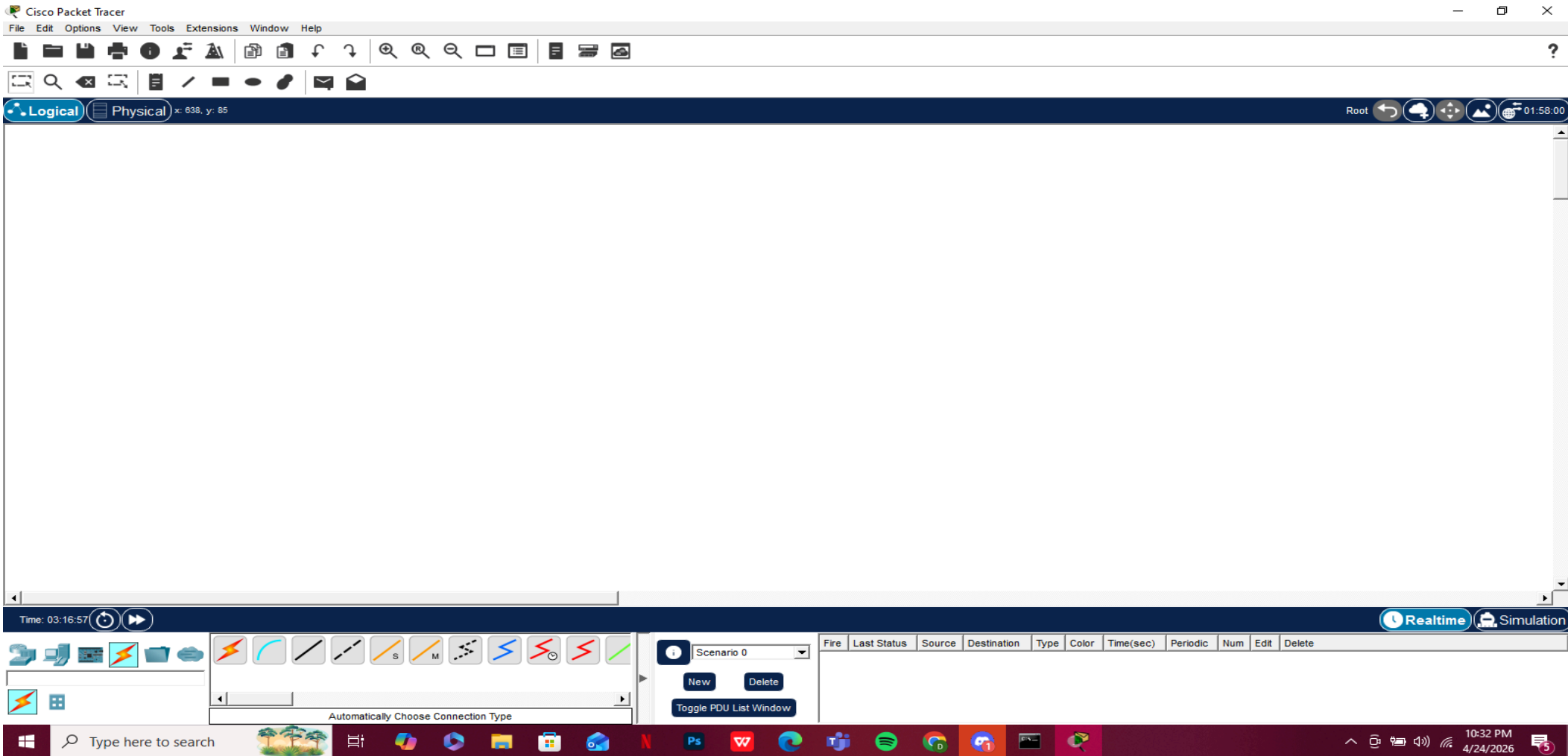
Enter valid login credentials such as email and password



Step 3:

Accessing the Packet Tracer Workspace.

After successful login, the Cisco Packet Tracer main workspace opens.



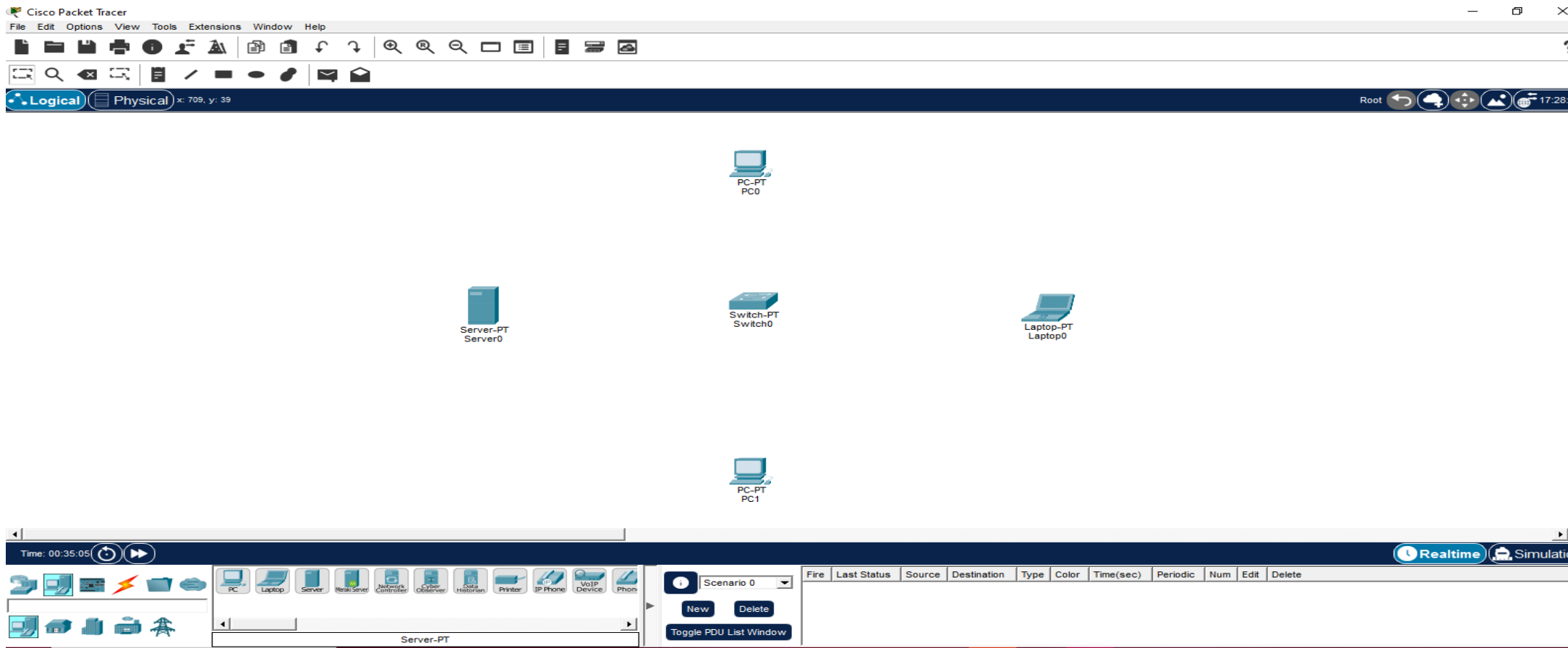
Step 4:

Adding Network Devices to the workspace

The required Network components (One DHCP Server, Two PCs, One Laptop, One Switch)

These Devices were accessed from the device-type selection panel located at the bottom of the interface.

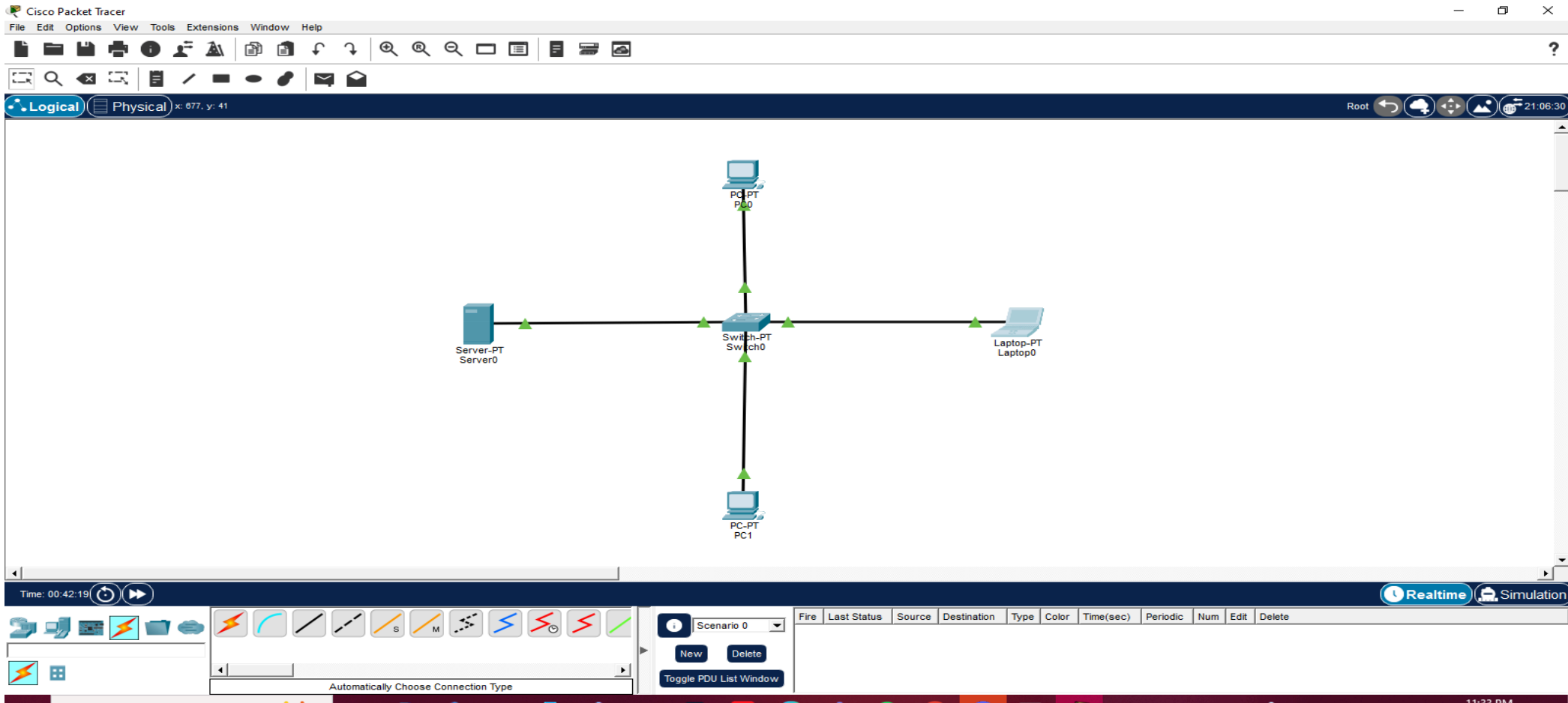
This is a Star topology.



Step 5:

Establishing Network Connection:

All devices(PCs, laptop and server) are successfully connected to the switch using copper straight-through cables. The green connection lines indicate that the links are active and communication has been fully established, not in the initialization stage. The switch acts as the central device in this star topology, allowing data to be transmitted between all connected devices.



Step 6:

Configuring the Server (DHCP Setup)

To assign static address to the server, click on the server icon > Click Desktop > click IP Configuration

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is shown with a central 'Switch-PT Switch0' connected to 'Server-PT Server0', 'PC-PT DAD', and 'PC-PT FORTH'. A red arrow labeled '1' points to the 'Server-PT Server0' icon. On the right, the 'Server0' configuration window is open, with the 'Desktop' tab selected. A red arrow labeled '2' points to the 'Desktop' tab, and another red arrow labeled '3' points to the 'IP Configuration' icon in the top-left corner of the Desktop window. The Desktop window shows various application icons such as 'IP Configuration', 'Terminal', 'Command Prompt', 'Web Browser', 'PC Wireless', 'VPN', 'AAA Accounting', 'Traffic Generator', 'MIB Browser', 'Cisco IP Communicator', 'Email', 'PPPoE Dialer', 'Text Editor', 'Firewall', and 'IPv6 Firewall'. The bottom status bar shows the time as 30:32:18.

Step 7:

Assigning Static IP Address to the Server:

The server is assigned a static IP address (172.50.35.0) to provide a stable connection before configuring the DHCP service

The screenshot displays a network simulation environment. On the left, a network diagram shows a central switch labeled 'Switch-PT Switch0' connected to three devices: 'Server-PT Server0' on the left, 'PC-PT DAD' at the top, and 'PC-PT FORTH' at the bottom. The 'Server0' configuration window is open on the right, with the 'Desktop' tab selected. The 'IP Configuration' section shows the 'Static' radio button selected, with the 'IPv4 Address' field set to '172.50.32.0'. The 'Subnet Mask' is '255.255.0.0', the 'Default Gateway' is '0.0.0.0', and the 'DNS Server' is '0.0.0.0'. The 'IPv6 Configuration' section shows the 'Static' radio button selected, with the 'IPv6 Address' field empty, the 'Link Local Address' set to 'FE80::2D0:BAFF:FE04:6512', and the 'Default Gateway' and 'DNS Server' fields empty. The '802.1X' section shows the 'Use 802.1X Security' checkbox unchecked, the 'Authentication' dropdown set to 'MD5', and the 'Username' and 'Password' fields empty. A red arrow points from the text '172.50.35.0' in the text block to the '172.50.32.0' in the 'IPv4 Address' field.

Server0

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.50.32.0

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::2D0:BAFF:FE04:6512

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Time: 30:49:59

Simulation

Step 8:

Configuring the DHCP Server Settings:

To begin, the DHCP icon was selected on the server to access the DHCP Configuration settings. The DHCP service was enabled on the server, the DHCP was configured starting from 172.50.35.20 within the 172.50.0.0/16 (Indicating class B IP address) Network. After entering the required settings, the configuration was saved.

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is shown with a central switch (Switch-PT Switch0) connected to a server (Server-PT Server0), a PC (PC-PT DAD), and another PC (PC-PT FORTH). The main window on the right shows the configuration for Server0, specifically the DHCP settings for the FastEthernet0 interface. The DHCP service is enabled, and a pool named 'serverPool' is configured with a start IP of 172.50.35.20 and a subnet mask of 255.255.0.0. The configuration is saved, and a table at the bottom lists the configured pool.

Server0 Configuration - DHCP Settings

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

Start IP Address: 172.50.35.20

Subnet Mask: 255.255.0.0

Maximum Number of Users: 512

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Configured Pools Table:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	172.50.35.20	255.255.0.0	512	0.0.0.0	0.0.0.0

Step 9:

To confirm the successful configuration of the DHCP server, each end device was tested for automatic IP address allocation.

All IP addresses were assigned dynamically without manual configuration, demonstrating that the DHCP service is functioning correctly. This ensures seamless communication among devices within the LAN network. This IP address obtained was within the 175.50.0.0/16.

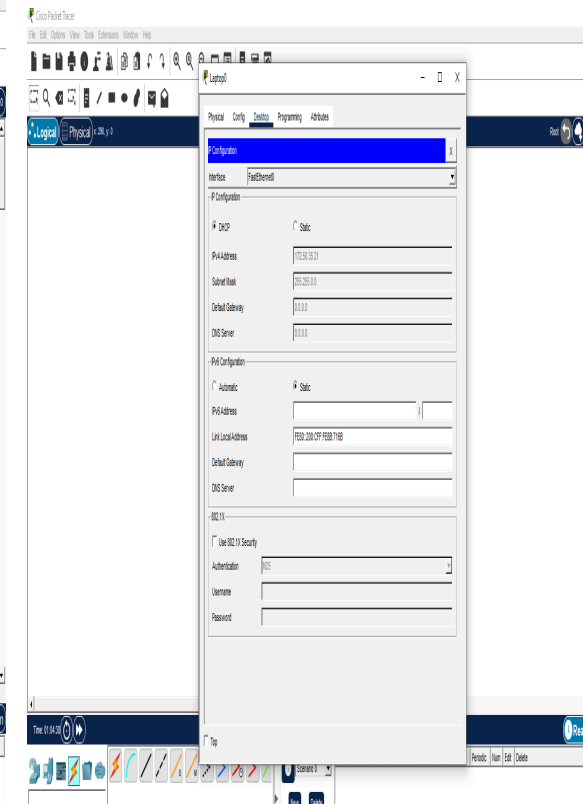
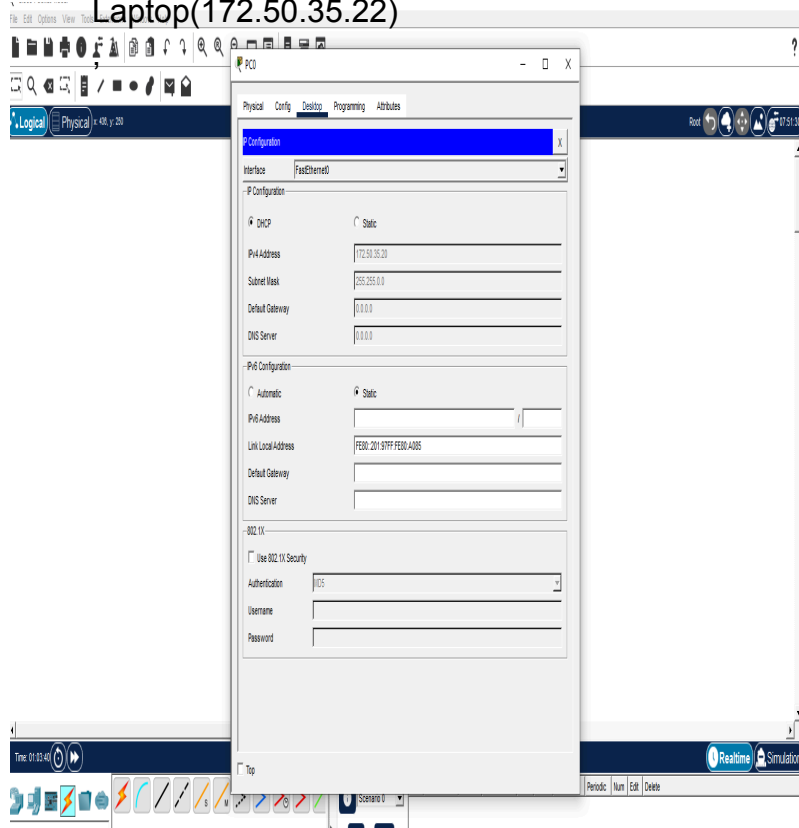
The first device(PC) successful
Obtained an IP address of 172.50.35.20,
Indicating proper communication with
the DHCP server
DAD's PC(172.50.35.20)

Laptop(172.50.35.22)

The second device (PC) successfully obtained an
IP address of 172.50.35.21, indicating proper
communication with the DHCP server

MUM's PC(172.50.35.21)

The first device(PC)
successfully obtained
an IP address of
172.50.35.22
Forth



Step 10:

Simple PDU Ping Test

After connection, a ping test was performed in simulation mode to verify network connectivity across all devices on the LAN. The results confirmed that each device could successfully communicate with others, indicating that the DHCP configuration and overall network setup were functioning correctly

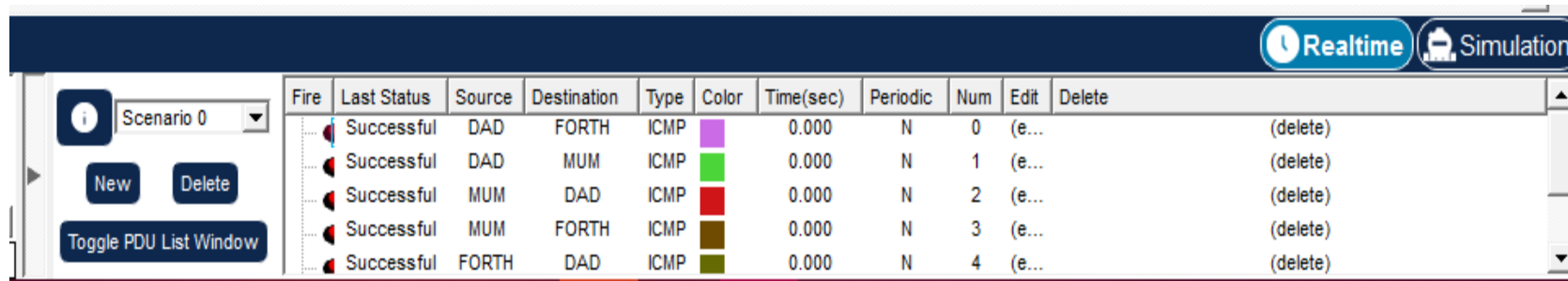
The image displays the Cisco Packet Tracer simulation environment. The network topology consists of a central Switch0 connected to four devices: Server-PT Server0, Laptop-PT MUM, PC-PT DAD, and PC-PT FORTH. A large red 'X' is drawn over the network diagram. The right-hand panel shows the 'Simulation Panel' with an 'Event List' table. The bottom status bar indicates the time is 32:28:18.335 and shows play controls. The bottom-right panel displays a table of events.

Vis.	Time(sec)	Last Device
...	7.827	Switch0
...	7.827	Switch0
...	9.824	--
...	9.825	Switch0
...	9.825	Switch0
...	9.825	Switch0
...	9.825	Switch0
...	11.822	--
...	11.823	Switch0
...	11.823	Switch0
...	11.823	Switch0
...	11.823	Switch0
...	13.820	--

Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	MUM	DAD	ICMP	Red	0.000	N	2	(e...)	(delete)
Successful	MUM	FORTH	ICMP	Red	0.000	N	3	(e...)	(delete)
Successful	FORTH	DAD	ICMP	Green	0.000	N	4	(e...)	(delete)
Successful	FORTH	MUM	ICMP	Blue	0.000	N	5	(e...)	(delete)

Conclusion

This project, Single LAN Setup successfully demonstrates the setup of a functional LAN with dynamic IP assignment using a DHCP server. All devices were able to automatically obtain valid IP addresses and communicate effectively, confirming a stable and correctly configured network. The simulation validates both connectivity and proper service deployment.



The screenshot shows a network simulation interface. At the top right, there are two buttons: 'Realtime' (with a clock icon) and 'Simulation' (with a server icon). On the left side, there is a control panel with a dropdown menu set to 'Scenario 0', 'New' and 'Delete' buttons, and a 'Toggle PDU List Window' button. The main area displays a table of traffic events.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
...	Successful	DAD	FORTH	ICMP	Purple	0.000	N	0	(e...	(delete)
...	Successful	DAD	MUM	ICMP	Green	0.000	N	1	(e...	(delete)
...	Successful	MUM	DAD	ICMP	Red	0.000	N	2	(e...	(delete)
...	Successful	MUM	FORTH	ICMP	Brown	0.000	N	3	(e...	(delete)
...	Successful	FORTH	DAD	ICMP	Dark Green	0.000	N	4	(e...	(delete)